

IS6200/IS6143 – Lab 1 Document

General guidelines

Naming

- When creating the filenames for your programs, please use the convention specified by your instructor. Otherwise, for programs that consist of just one file, name the file *first_last_program.py* where *first_last* specifies your first and last name and *program* specifies the name of the program. For programs that have multiple files, store the files in a folder named *first_last_program*.
- When creating names for variables and functions, please use the guidelines and recommendations specified by your instructor.

User interfaces

- You should think of the user interfaces that are shown for the projects as starting points. If you can improve on them, especially to make them more user-friendly, by all means do so.

Specifications

- You should think of the specifications that are given for the projects as starting points. If you have the time to enhance the programs by improving on the starting specifications, by all means do so.

Top-down development

- Always start by developing a working version of the program for a project. That way, you'll have something to show for your efforts if you run out of time. Then, you can build out that starting version of the program until it satisfies all of the specifications.
- You should use top-down coding and testing as you develop your programs. You might also want to sketch out a hierarchy chart for each program as a guide to your top-down coding.

Files supplied by your instructor

- Some of the projects require starting text, CSV, or binary files. These files are identified in the specifications for the projects, and your instructor should make these starting files available to you.

Project 1-1: Student Registration

Create a program that allows a student to complete a registration form and displays a completion message that includes the user's full name and a temporary password.

Console

```
Registration Form

First name:      Eric
Last name:       Idle
Birth year:      1934

Welcome Eric Idle!
Your registration is complete.
Your temporary password is: Eric*1934
```

Specifications

- The user's full name consists of the user's first name, a space, and the user's last name.
- The temporary password consists of the user's first name, an asterisk (*), and the user's birth year.
- Assume the user will enter valid data.

Project 1-2: Pay Check Calculator

Create a program that calculates a user's weekly gross and take-home pay.

Console

```
Pay Check Calculator

Hours Worked:    35
Hourly Pay Rate: 14.50

Gross Pay:       507.5
Tax Rate:        18%
Tax Amount:      91.35
Take Home Pay:   416.15
```

Specifications

- The formula for calculating gross pay is:
`gross pay = hours worked * hourly rate`
- The formula for calculating tax amount is:
`tax amount = gross pay * (tax rate / 100)`
- The formula for calculating take home pay is:
`take home pay = gross pay - tax amount`
- The tax rate should be 18%, but the program should store the tax rate in a variable so that you can easily change the tax rate later, just by changing the value that's stored in the variable.
- The program should accept decimal entries like 35.5 and 14.25.
- Assume the user will enter valid data.
- The program should round the results to a maximum of two decimal places.

Project 1-3: Tip Calculator

Create a program that calculates the tip and total for a meal at a restaurant.

Console

```
Tip Calculator

Cost of meal: 52.31
Tip percent: 20

Tip amount:    10.46
Total amount: 62.77
```

Specifications

- The formula for calculating the tip amount is:
$$\text{tip} = \text{cost of meal} * (\text{tip percent} / 100)$$
- The program should accept decimal entries like 52.31 and 15.5.
- Assume the user will enter valid data.
- The program should round the results to a maximum of two decimal places.

Project 1-4: Price Comparison

Create a program that compares the unit prices for two sizes of laundry detergent sold at a grocery store.

Console

```
Price Comparison

Price of 64 oz size: 5.99
Price of 32 oz size: 3.50

Price per oz (64 oz): 0.09
Price per oz (32 oz): 0.11
```

Specifications

- The formula for calculating price per ounce is:
$$\text{price per ounce} = \text{price} / \text{ounces}$$
- Assume the user will enter valid data.
- The program should round the results to a maximum of two decimal places.

Project 1-5: Travel Time Calculator

Create a program that calculates the estimated hours and minutes for a trip.

Console

```
Travel Time Calculator

Enter miles: 200
Enter miles per hour: 65

Estimated travel time
Hours: 3
Minutes: 5
```

Specifications

- The program should only accept integer entries like 200 and 65.
- Assume that the user will enter valid data.

Hint

- Use integers with the integer division and modulus operators to get hours and minutes.